

SHARWIL BHENDE

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ABOUT

Computer Engineering student at Symbiosis Institute of Technology with a strong foundation in problem-solving, data structures, and databases, and hands-on exposure to AI/ML. Experienced in building and deploying real-world applications end-to-end using Python and SQL, with a focus on creating practical, impactful solutions.

SKILLS

Languages: Python, C/C++, SQL

Core Areas: Machine Learning, Deep Learning, SQL, DBMS, Object-Oriented Programming (OOP), Data Structures & Algorithms

Core Strengths: Problem Solving, Analytical Thinking, Adaptability & Fast Learning, Strong Communication, Team Collaboration

EDUCATION

B.Tech, Computer Engineering

2023–2027

Symbiosis Institute of Technology, Nagpur

PROJECTS

Real-Time Indian Sign Language Recognition

- Building a real-time sign language recognition system for the Deaf and hard-of-hearing community using MediaPipe Holistic and a GRU classifier.
- Architecting an agentic fallback layer with Gemini function-calling for robust sentence assembly on low-confidence predictions.
- Collecting a real, signer-independent dataset in collaboration with a partner organization supporting the Deaf and hard-of-hearing community.
- Targeting a Q1 journal publication for this deep learning-based sign language detection research.

Adaptive Traffic Signal Control (YOLOv8-based)

- Built a comparative study of vehicle-detection models for dynamic, density-based signal timing.
- Evaluated tradeoffs across detection models to optimize real-time traffic flow.
- Findings presented at IICTDS 2025, NMIMS Chandigarh.

Smart Civic Portal — AI-Powered Civic Governance System

- Solo-built full-stack platform (Flask, MongoDB, Gemini API) for citizen-government interaction.
- Integrated Gemini API for AI-assisted issue classification and response support.
- Designed and deployed end-to-end, from database schema to production-ready system, as a solo developer.

CropCure — AI-Based Cotton Leaf Disease Detection App

- Built an AI-powered mobile app using MobileNetV2 to detect Cotton Leaf Curl Disease (CLCuD) with confidence-score outputs.
- Achieved 94.8% classification accuracy; findings presented at IC3I Conference.
- Designed for real-world agricultural use to support crop yield improvement.

RESEARCH & PUBLICATIONS

- “Enhancing Cotton Yield Through Detection of Cotton Leaf Curl Disease” — Presented at IC3I Conference.
- “Comparative Study of YOLOv5, YOLOv8, Faster R-CNN, SSD for Traffic Detection” — Presented at IICTDS 2025.
- Deep learning-based Indian Sign Language detection — targeting Q1 journal publication.

EXTRACURRICULAR

- **Secretary-General, Model United Nations** — led and coordinated an event of 100+ participants.
- Represented as a delegate across multiple committees, developing strong negotiation, diplomacy, and public speaking skills.

ACHIEVEMENTS

- **Shri C. Subramaniam Award for Excellence in Character**, Bharatiya Vidya Bhavan — recognized for leadership, integrity, and strong personal values.
- **Top Performer**, Industry Readiness Program 2025–2026.